Do Now · retrieval

Programming Techniques · Sequence and Selection

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What does a full adder include that a half adder does not?

Q2

What is a local variable?

Q3

Why does a stack use LIFO?

Q4

Simplify A ∨ (A ∧ B).

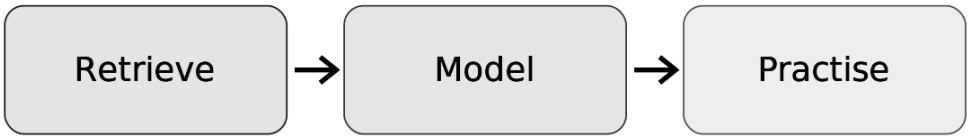
New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Explain the important decision — Sequence changes score before selection tests it. The condition is true at exactly 10 because >= includes the boundary.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

sequence — Instructions run in order

selection — A conditional branch

Boolean — True or false expression

condition — A logical test

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

Follow the order of execution

Follow the worked method

1. Update in sequence — score starts at 7. Adding 3 changes the same variable to 10; the condition has not been tested yet.

2. Evaluate the condition — Substitute the new value: 10 >= 10 is true. Using the old value 7 would trace the wrong state.

3. Take the selected branch — The true branch prints Pass. For initial scores 6, 7 and 8, the updated scores are 9, 10 and 11: only the latter two print Pass.

Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Discuss, then respond
value = 3; value *= 4; is value > 10?

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Apply independently

Trace x = 4; x = x + 7; output x. [2]

Question 2

Apply independently

Write a C# condition that prints Adult when age >= 18. [3]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Apply independently

Explain one difference between sequence and selection. [2]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

True or false: an else branch runs when its if condition is false.

Use key vocabulary from memory

1. Explain sequence in your own words.

2. Explain selection in your own words.

3. Explain Boolean in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.